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SCIENCE DIRECT
Precambrian Research 142 (2005) 83–92
Precambrian Research
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Paleomagnetism of the Baiyisi volcanic rocks (ca. 740 Ma) of Tarim, Northwest China: A continental fragment of Neoproterozoic Western Australia?

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Received 19 January 2005; received in revised form 15 June 2005; accepted 13 September 2005

Abstract

We report a paleomagnetic pole from the SHRIMP U-Pb zircon dated Baiyisi volcanic rocks in the Xishankou area at the northern edge of the Tarim Block, Northwest China. Rock magnetic and microscopic observations indicate that the characteristic remanent magnetization (ChRM) isolated by both thermal and alternating-field demagnetization is preserved in original single domain (SD) or pseudo-single domain (PSD) low-titanium titanomagnetite grains. The new paleomagnetic pole ($Q = 5$) indicates that the Tarim Block was located in equatorial paleolatitudes at ca. 740 Ma and was most likely situated against Western Australia during the most of Neoproterozoic and the early Paleozoic. We suggest that the Tarim Block could be a continental fragment rifted from Western Australia during the break-up event of the Rodinia supercontinent at ca. 820–750 Ma.
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Keywords: Tarim Block; Baiyisi volcanic rocks; Rodinia; Paleomagnetism; Neoproterozoic

1. Introduction

Following Hoffman's (1991) earlier Neoproterozoic reconstruction of the Rodinia supercontinent proposing a connection between the East and Southeast Asian blocks and Northern Australia, cumulative studies have shown that most Eastern Asian blocks were not juxtaposed to Northern Australia during the Neoproterozoic (e.g. Li et al., 1995, 1996, 2004; Li and Powell, 2001; Piper and Zhang, 1997; Zhang and Piper, 1997; Evans et al., 2000; Yang et al., 2004). For the Tarim Block of central Asia (Fig. 1a), the connection of Tarim to Northern Australia has remained unchanged in Neoproterozoic reconstructions for more than a decade (e.g. Hoffman, 1991; Li et al., 1996, 2003b; Powell and Pisarevsky, 2002; Chen et al., 2004). Notwithstanding this, the paleoposition of the Tarim Block is uncertain for most of the Neoproterozoic. The putative paleogeographic position of Tarim in Neoproterozoic reconstructions is generally based on (1) biostratigraphic and tectonostratigraphic correlation of Neoproterozoic successions between the Tarim Block and Australia (e.g. Brookfield, 1994; Li et al., 1996) and (2) a close early Paleozoic biogeographic and lithostratigraphic association of Tarim with the South China

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doi:10.1016/j.precamres.2005.09.006

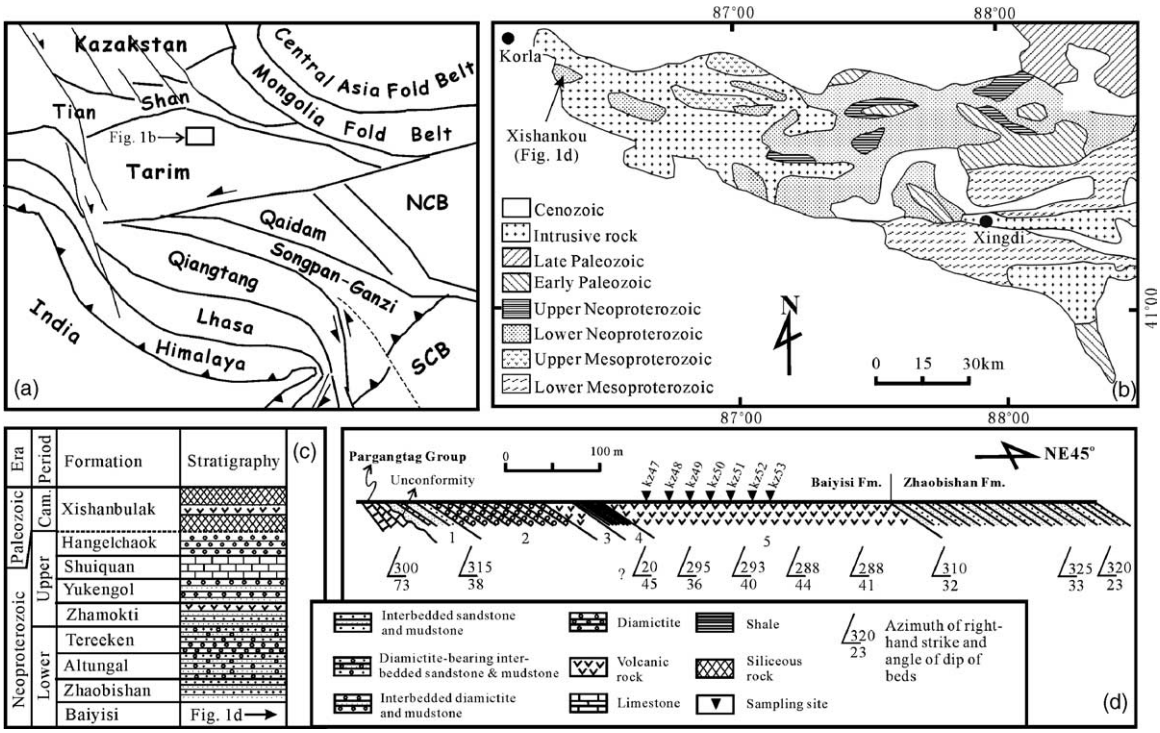


Fig. 1. (a) Sketch of distribution of the major tectonic units in the western China and its surroundings. (b) Simplified map of the northeastern Tarim Block showing distribution of the Neoproterozoic strata and the sampling location after Xu et al. (2005). (c) Neoproterozoic stratigraphic column in the Quruktagh area of Tarim. (d) Succession of the Neoproterozoic Baiyisi Formation in the Xishankou area showing paleomagnetic sampling details as well as the bedding attitudes of the Baiyisi Formation and its underlying and overlying strata.

Block (SCB) and Australia (e.g. Burrett and Richardson, 1980; Burrett et al., 1990; Nie and Rowley, 1991; Shu and Chen, 1991). Although the Tarim Block is known to incorporate a complete mid- and late-Neoproterozoic succession, including several horizons of rift volcanics and three glaciations (e.g. Gao and Qian, 1985; Gao et al., 1985; Brookfield, 1994; Xiao et al., 2004; Xu et al., 2005), and some new geochronologic and geochemical data provide constraints on the ages of the glaciations and the correlation with glaciations elsewhere (Xu et al., 2003, 2005; Xiao et al., 2004), few reliable paleomagnetic data have been reported to help for constrain the geodynamics of the Tarim Block during the Neoproterozoic.

Recently, Chen et al. (2004) reported a paleomagnetic pole from the 807 ± 12 Ma Aksu dyke swarm (ADS) in the Aksu area in the northwestern Tarim Block, and proposed that Tarim might have been juxtaposed to the north of Australia with the Aksu dykes being the northward continuation of lamprophyre dykes and kimberlite pipes in northeast Kimberley district of Western Australia. This ca. 800 Ma paleomagnetic pole provided firsthand evidence for the Kimberley–Tarim connection (Li et al., 1996); however, the small angular differences between the in situ site-mean direction for the ADS (Chen et al., 2004) and both the present earth field and geomagnetic dipolar field leaves some doubt about its reliability as a primary paleomagnetic pole. A variant of the revised Kimberley–Tarim connection (Chen et al., 2004) places the present northern edge of the Tarim Block offshore rather than facing Australia. Since the thickest and most complete mid- and late-Neoproterozoic successions were developed in the Quruktagh region at the northern edge of the Tarim Block (Fig. 1a and b; Gao and Qian, 1985; Gao et al., 1985) and successive rifting is necessary to preserve such complete successions (Brookfield, 1994), the revised Kimberley–Tarim connection (Chen et al., 2004) could have been a short-lived connection and may have been disaggregated in the break-up of the early Neoproterozoic supercontinent at ca. 820–750 Ma (Li et al., 2003b).

In this paper, we present a paleomagnetic result from the SHRIMP U-Pb zircon dated Neoproterozoic Baiyisi volcanic rocks in the Xishankou area at the northern edge of the Tarim Block, which aims to help us understand the paleogeography of the mid- and late-Neoproterozoic glaciations of the Tarim Block, and permit a paleomag-

netic test for the configuration of Tarim in Neoproterozoic continental reconstructions.

2. Geology and paleomagnetic sampling

The northern edge of the Tarim Block is well-known for a thick and almost complete mid- and late-Neoproterozoic succession characterized by several horizons of glacial deposits and rift volcanic rocks separated by thick deposits predominantly of shales, sandstones and limestones (Fig. 1b; Gao and Qian, 1985; Gao et al., 1985; Brookfield, 1994; Xiao et al., 2004; Xu et al., 2003, 2005). According to the correlation based mainly on lithology, the Neoproterozoic succession is divided into lower and upper parts and includes eight formations (Gao and Zhu, 1984; Gao and Qian, 1985), possibly with an unconformity between the lower and upper series (e.g. Wang, 1986). The lower part unconformably overlies the Mesoproterozoic Pargangtag Group and includes, in ascending order, the Baiyisi, Zhaobishan, Altungal and Tereeken formations, whereas the upper part is unconformably overlain by lower Cambrian siliceous rocks (the Xishanbulak Formation) and is divisible into the Zhamokti, Yukengol, Shuiquan and Hangelchaok formations (Fig. 1c).

We have sampled the Baiyisi volcanic rocks from the Xishankou section (41.60°N, 86.54°E) at the western end of the Quruktagh area, northern Tarim (Fig. 1a and b), where three Neoproterozoic formations, Baiyisi, Zhaobishan and Altungal could be identified. The Baiyisi Formation in Xishankou has a thickness of about 640 m and consists mainly of five layers (Fig. 1d) comprising about 13 m of interbedded layer of sandstones and mudstones at the base then about 177 m of diamictites followed by ~23 m of volcanic rocks, ~23 m of shales and ~404 m of volcanic rocks at the top. The SHRIMP U-Pb age analysis gives a weighted mean age of 755 ± 15 Ma for sample (Xb006) collected near the base of the Baiyisi volcanic rocks in the southeast of the Xinger area (Xu et al., 2005) and indicates the oldest eruption age of the Baiyisi volcanic rocks. The Baiyisi volcanic rocks in Xishankou that appear above the Baiyisi diamictite unit (Fig. 1d; Xiao et al., 2004) have also been geochronologically studied and a SHRIMP U-Pb zircon age of 727 ± 10 Ma has been assigned to the top horizon of volcanic rocks in the Baiyisi Formation (Li Z.X., personal communication). Geochemical studies suggest that the Baiyisi volcanic rocks in the Xishankou, Saimashan and Xinger areas of Quruktagh are made up of a bimodal volcanic suite comprising basalt and dacite-rhyolite, and resembling the tholeiitic basalts in continental rift (Gao et al., 1985; Jiang et al., 2001; Xu et al., 2003, 2005). Altogether seven sites were sampled in basalts using a portable gasoline-powered drill and oriented by sun compass. For each site, between 8 and 10 cores were generally collected from discrete basalt flows, however, because the portable drill broke down in the field, only 4 cores were collected for the site kz53. The bedding attitude of the Baiyisi volcanic rocks (Strike/Dip = 305°/37°, α_{95} = 9.5°) was determined from eight individual measurements from the spiracles of volcanic rocks and both the underlying and overlying deposits with the exclusion of an anomalous and perhaps questionable measurement from the underlying shales (Strike/Dip = 20°/45°), which shows ca. 90° difference in azimuth of strike with the other observations (Fig. 1d).

3. Results

Following sample preparation one pilot 2.2-cm long cylindrical specimen from each site was subjected to acquisition and back-demagnetization of isothermal remanent magnetization (IRM) using JR-5A spinner magnetometer and 2G660 pulse magnetizer. These experiments show saturation to occur by 0.3–0.5 T with remanence coercivity (Hcr) deduced from back-demagnetization of the saturation IRM to be less than 50 mT (Fig. 2). These results indicate low-coercivity magnetite or low-titanium titanomagnetite as the predominant carrier of remanence in the Baiyisi volcanic rocks.

A total of 51 specimens were subsequently subjected to either progressive thermal or alternating field (AF) demagnetization. Progressive thermal demagnetization up to 550–590 °C was carried out in 17–18 steps in a MMTD60 thermal demagnetizer with a residual magnetic field of less than 10 nanotesla (nT) in the center of the oven. Progressive AF demagnetization up to 80–90 mT was conducted in 12–13 steps using an AF

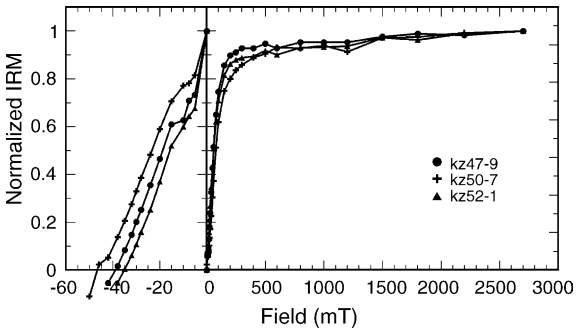


Fig. 2. Behavior of the acquisition and back-demagnetization of isothermal remanent magnetization (IRM).

demagnetizer 2G600 coupled to a 2G-755R cryogenic magnetometer. All measurements of natural remanent magnetization (NRM) were made on a 2G-755R cryogenic magnetometer inside a field-free space (<300 nT) in the Paleomagnetism Laboratory of the Institute of Geology and Geophysics, Chinese Academy of Sciences. Results were analyzed by principal component analysis (Kirschvink, 1980) and Fisher statistics (Fisher, 1953).

Initial magnetic intensities of the specimens vary from 6.02×10^{-2} to 2.12 mA/m with an average value of 0.433 mA/m. Thermal and AF demagnetization defines two NRM components. A component A is isolated below peak demagnetization temperature of 330 °C or peak alternating field of 20–25 mT (Fig. 3). The component A is mostly of normal polarity with only one specimen showing reversal polarity. Directions of this first-commonly removed component are well clustered and have an in situ mean direction of $D = 30.5°$, $I = 60.3°$ with $\alpha_{95} = 3.8°$ (48 out of 51 specimens demagnetized), which is close to the present geomagnetic field ($D = 2°$, $I = 61°$) suggesting a recent remagnetization of viscous origin (Fig. 4a and b). A different component B decaying almost linearly to the origin is defined up to peak demagnetization temperatures 510–530 °C in general (Fig. 3a, c, e and f) and to as high as 570 °C in a few specimens (Fig. 3b) or peak alternating fields of 70–90 mT (Fig. 3d). This characteristic remanent magnetization (ChRM) component is of single polarity. All specimens demagnetized, except eight (Table 1) that show either unstable higher temperature demagnetization behavior or great deviation from site observations, yield accurately defined ChRM directions with the maximum angular deviation (MAD) of less than 10°. The site-mean directions of the component B are also well grouped and yield a mean direction of $D = 250.4°$, $I = -29.7°$ before and $D = 245.2°$, $I = 1.7°$ and $\alpha_{95} = 6.0°$ after, tilt correction (Table 1; Fig. 4c and d). A fold test is not possible because the bedding dips uniformly (Fig. 1d). The *F*-statistics (McFadden and Lowes, 1981) of site-mean directions shows that directions of site kz47 and kz52 + 3 are significant different from each other and from the other site-mean directions, indicating that the overall mean is an average of at least three different spot readings of the geomagnetic field. This overall mean has a dispersion of 7.3°, which is some 5° smaller than that predicted for paleolatitudes less than 10° for the past 5 Ma (Merrill and McElhinny, 1983). However, a low reversal frequency of the geomagnetic field in Precambrian time (Dunlop and Yu, 2004) could be the cause of smaller dispersion than in recent times.

Table 1
Characteristic directions of the Neoproterozoic Baiyisi volcanic rocks from the Xishankou area (41.60°N, 86.54°E), Tarim

Site ID	n/n_0	Dg	Ig	Ds	Is	k	α_{95}
kz47	8/8	243.3	−40.4	236.3	−6.6	48.8	8.0
kz48	8/8	251.9	−29.6	245.5	1.4	65.9	6.7
kz49	6/8	252.7	−29.0	247.3	1.7	68.6	8.1
kz50	7/8	250.7	−27.5	246.2	3.7	42.3	9.4
kz51	8/8	253.3	−31.7	246.8	−1.0	46.1	8.2
kz52 + 3	6/11	249.6	−20.0	247.9	11.1	48.5	9.7
Mean	(6/6)	250.4	−29.7	245.2	1.7	124.7	6.0

Abbreviations: Site ID, site identification; n/n_0, number of specimens used in calculation/number of specimens demagnetized; k and α_{95}, precise parameter of Fisher distribution and half-cone-angle of the 95% confidence circle; Dg/Ig (Ds/Is), declination/inclination in geographic (stratigraphic) coordinate.

4. Discussion

4.1. Origin of the characteristic remanence

Both thermal and alternating-field demagnetization were able to isolate a characteristic remanent magnetization up to peak demagnetization temperatures 510–570 °C in the former case or peak alternating fields of 70–90 mT in the latter case. Together with the magnetic behavior of acquisition and back-field demagnetization of the saturation IRM, this implies that the ChRM component is most likely to be carried by low-titanium titanomagnetite. The thermomagnetic analysis of magnetic susceptibility using a CS-3 apparatus coupled to KLY-3 Kappabridge indicates a reversibility of the heating and cooling curves before ca. 430 °C and a rapid increase in magnetic susceptibility around ca. 500 °C (Fig. 5) that might be evidence for thermal mineralogical alteration of titanomagnetite to magnetite. Scanning electronic microscopy (SEM) on pilot samples showed that the magnetic grains are of 0.1–0.2 μm single domain (SD) or pseudo-single domain (PSD) size (Fig. 6a and b) and are interspersed in a matrix groundmass of basalt, and some are tear-shaped (Fig. 6a and b), suggesting a typical rapidly cooled texture of volcanic rocks. Moreover, microscopic observation through polished thin sections indicates an essentially original magmatic texture, with no chemical alteration caused by the fluid or weathering in the groundmass (Fig. 6c and d). Therefore, the magnetic grains of our samples are most likely to be the original ones so that the ChRM carried by them could be the primary magnetization. Given that directions pointing to southwest with shallow inclinations these are of normal polarity by comparison with the present day field although the true polarity in Neoproterozoic times is of course,

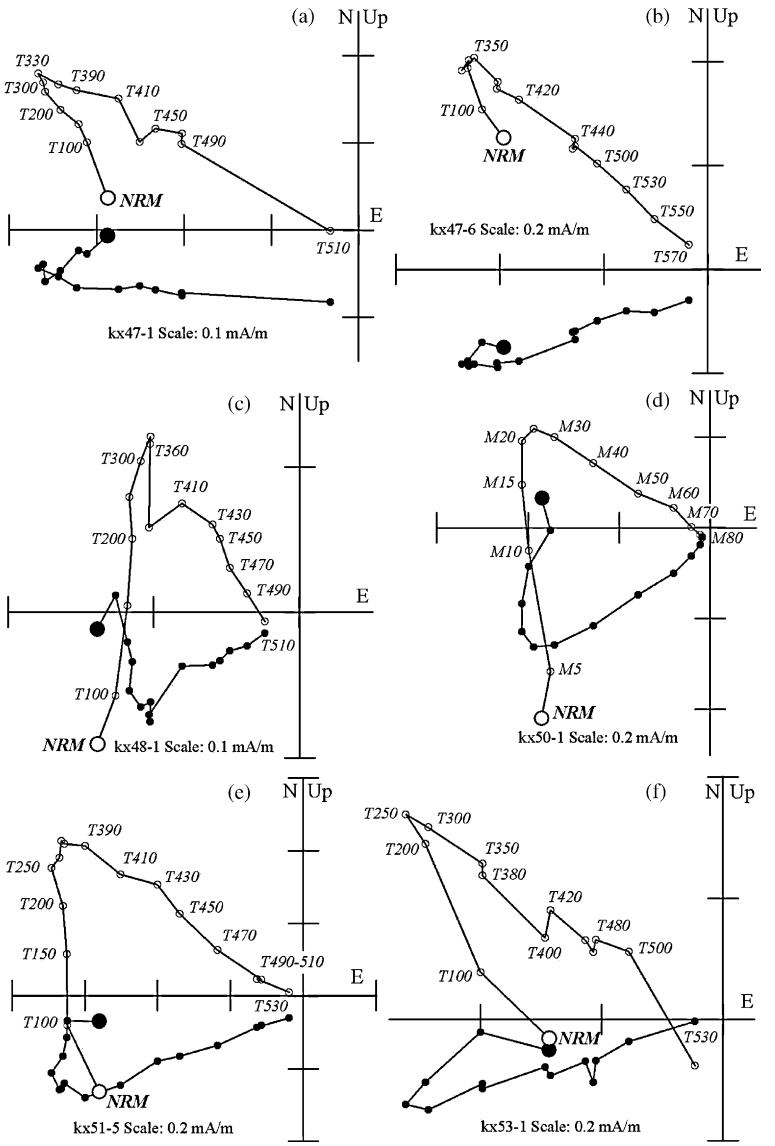


Fig. 3. Typical orthogonal vector plots. Directions are plotted in situ; solid/open circles represent vector endpoints projected onto the horizontal/vertical plane; *T* and *M* imply thermal (°C) and alternating-field (mT) demagnetization, respectively.

unknown. The corresponding paleomagnetic pole of the ChRM lies at 14.2°E, 17.7°S with $d_p/d_m = 3.0°/6.0°$ and yields a paleolatitude of $0.9 \pm 3.0°$N. This paleopole (as well as the antipole) are significantly different from both the preliminary mid- and late-Neoproterozoic mean pole from the northwestern margin of Tarim (Li et al., 1991) and from the Phanerozoic apparent polar wander paths for the Tarim Block (e.g. Li, 1990; Fang et al., 1998), indicating that the new paleopole is dissimilar to any known younger paleopoles for Tarim. In terms of *Q*-factor (van der Voo, 1990) the new paleopole for the Baiyisi volcanic rocks (ca. 740 Ma) is ranked as $Q = 5$, and therefore, well able to be used for inferring the Neoproterozoic paleogeographic reconstruction for Tarim.

4.2. Neoproterozoic paleogeography of Tarim

There is remarkable similarity in the stratigraphic sequence, the ages of the Neoproterozoic thermotectonic events, and in the ages of cratonization between Tarim and South China (Lu et al., 2003 and references therein). Hence, it is likely that the Tarim Block might have been juxtaposed against the SCB during the most of Neoproterozoic times. The debate regarding the South China paleoposition within the Neoproterozoic super-

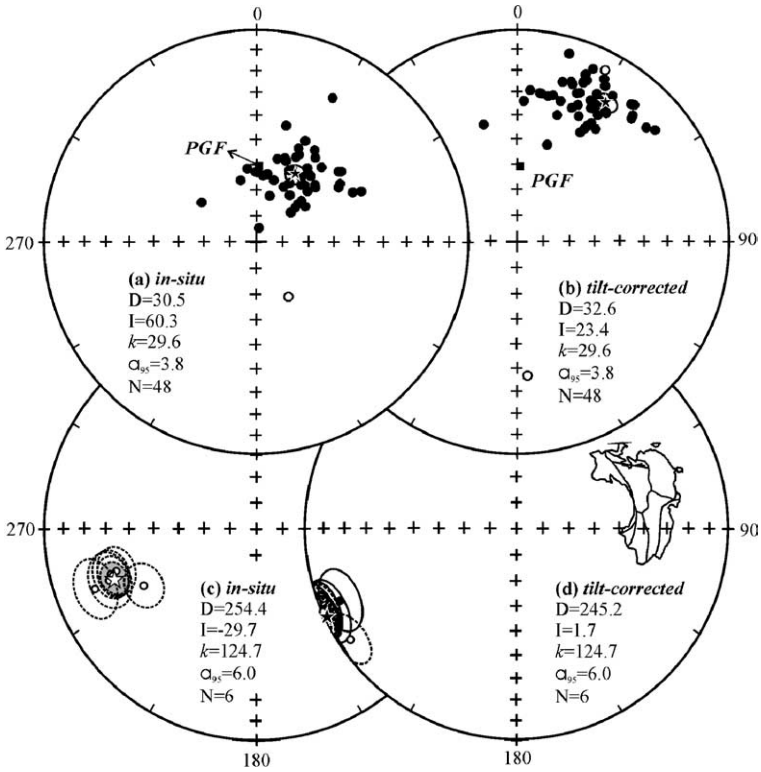


Fig. 4. Equal-area projections of (a and b) directions of the component A and (c and d) site-mean directions of characteristic component B before and after tilt adjustment. Solid/open symbols represent downward/upward inclinations; stars indicate the overall-mean directions with 95% confidence limits. PGF implies present geomagnetic field.

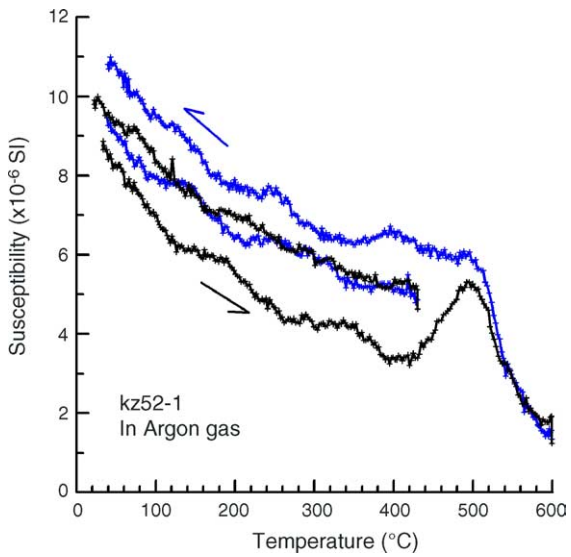


Fig. 5. Progressive temperature dependence of magnetic susceptibility (k–T curve) of representative sample during a heating-cooling cycle in argon gas atmosphere.

continent and its relationship to Tarim is evaluated within this general constraint (e.g. Li et al., 1995, 2003b; Li and Powell, 2001; Zhao et al., 1996; Zhang and Piper, 1997; Evans et al., 2000; Yang et al., 2004). One current debate regarding the paleoposition of South China in the early Neoproterozoic Rodinia supercontinent is the 'missing link' model (Li et al., 1995, 2003b; Li and Powell, 2001) and the South China–Northwestern Australia connection (e.g. Zhao et al., 1996; Zhang and Piper, 1997; Evans et al., 2000; Yang et al., 2004; Zhang S.H., personal communication). We note that an early Neoproterozoic sedimentary hiatus is common in both South China and Australia (e.g. Wang, 1986; Li et al., 1996; Preiss, 2000; Wang and Li, 2003), and direct paleomagnetic constraints for their paleopositions in the early Neoproterozoic supercontinent are therefore impossible. The 'missing link' model has to be based mainly on the tectonostratigraphic correlations (Li et al., 1995, 2003b; Li and Powell, 2001). However, an overall weighted mean mid-Neoproterozoic paleomagnetic pole (04.4°N, 161.1°E, A_{95} = 12.9°, Q = 7) for the SCB at 748 ± 12 Ma (Zhang and Piper, 1997; Evans et al., 2000) suggests that the SCB was located around 30–40° and adjacent

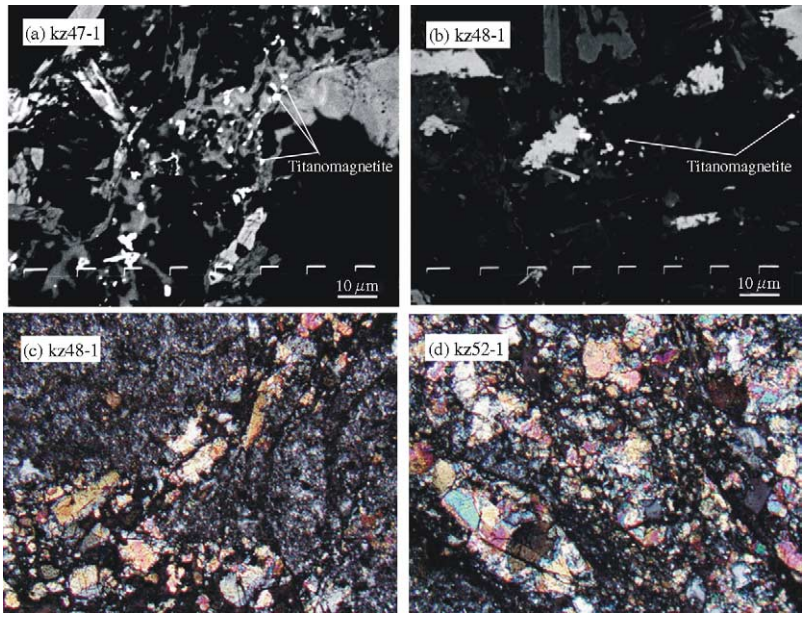


Fig. 6. (a and b) Scanning electronic microscopy (SEM) and (c and d) microscopic observation results for pilot samples from the Baiyisi volcanic rocks.

to Northwestern or Northeastern Australia (Fig. 13 of Evans et al., 2000). Comparison of the APWP from the SCB and that of Australia further supports the view that the SCB may have been located adjacent to Northwestern Australia from ca. 750 Ma until the early Paleozoic (Yang et al., 2004; Zhang S.H., personal communication). It is therefore possible that the SCB moved from southeast to northwest to Australia before ca. 750 Ma. Alternatively, as the original author of the 'missing link' model recently favored, the SCB may have been the vicinity of Western Australia during ca. 800–750 Ma (Fig. 5 of Li et al., 2004). The South China–Northwestern Australia connection can also account for the comparable Neoproterozoic tectonostratigraphic records in South China and Australia and it is probably better than the 'missing link' model for explaining the first episode of major plume activity in Rodinia between 830 and 795 Ma, which was prominent in South China, India, Australia and southern Africa but when there was little activity in Laurentia (Li et al., 2003b, 2004). In addition, Yang et al. (2004) also proposed that the long-lived paleomagnetic fit between the SCB and Australia from ca. 750 Ma to the early Paleozoic might have started prior to ca. 1 Ga, thus allowing a continuation of the diachronic orogenic belts between the Jiangnan Grenvillian belt and the Rudall belt or Miles orogeny in Western Australia. Hence, the SCB was most likely located adjacent to Northwestern Australia during the most of Neoproterozoic times.

Given that the South China–Northwestern Australia fit was the case at ca. 800 Ma, we speculate a relative paleoposition of Tarim relative to South China and Australia before the break-up of the Rodinia supercontinent (position 0 in Fig. 7, dashed line), in which Tarim was possibly juxtaposed to both South China and Australia

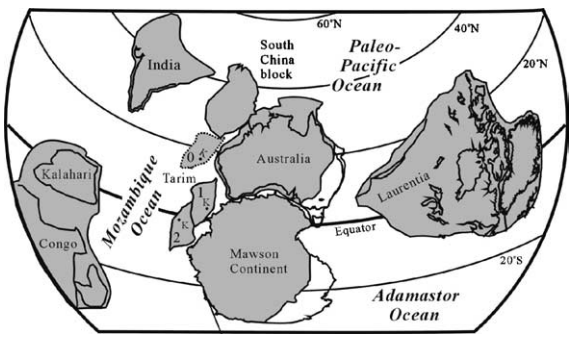


Fig. 7. A tentative Neoproterozoic paleogeographic reconstruction of the Tarim Block with respect to west Rodinia at ca. 740 Ma in the geographic coordinates. South China is shown according to Yang et al. (2004) and the others are shown after Torsvik (2003), Meert (2003) and Meert and Torsvik (2003). Relative longitudes are unconstrained in the reconstruction. K stands for the Quruktagh area of the northeastern Tarim Block. A conjectural paleoposition of Tarim in the early Neoproterozoic Rodinia supercontinent relative to Australia and South China is shown as position 0 (dashed line). Possible paleoposition of Tarim at ca. 740 Ma (position 1 or 2) is shown after superposing our ca. 740 Ma paleopole or its antipole and the Mundine Well dyke swarm (MDS) paleopole from Australia (Wingate and Giddings, 2000).

with its present northwestern and southwestern margins against western South China and Western Australia. This configuration allows a connection between the newly reported Grenvillian orogeny (ca. 1.0 Ga) in the southwestern margin of the Tarim Block (Zhang et al., 2003a) and the Grenvillian Sibao Orogeny in the central of South China (Li et al., 1995, 2002). It also allows a correlation of magmatism around 800 Ma between both northwestern and southwestern margins of Tarim (Zhang et al., 2003b, 2004; Zhang et al., in revision; Chen et al., 2004; Zhang L.F., unpublished data) and the SCB (Li et al., 2003a, 2003b).

The new ca. 740 Ma paleomagnetic pole indicates that the Tarim Block was located in equatorial paleolatitudes supportive of low latitudes for the Sturtian glaciations (e.g. Evans, 2000) and for the hypothesis of a snowball earth (e.g. Hoffman et al., 1998) at ca. 750–720 Ma. Because of uncertainty of polarity of this new Precambrian paleopole, Tarim may have been located adjacent to Southwestern Australia (positions 1 and 2 in Fig. 7) with either slight counterclockwise or large clockwise rotation with respect to Tarim's conjectural paleoposition in the early Neoproterozoic (position 0 in Fig. 7, dashed line). However, we prefer to place the Tarim Block in the vicinity of the Southwestern Australia during ca. 740 Ma with the now north-facing Neoproterozoic to Devonian passive margin facing the Australia and Mawson continents (position 1 in Fig. 7). According to this configuration, the Tarim Block was subjected to a southward drift with a slight counterclockwise rotation with respect to its conjectural early Neoproterozoic configuration after the occurrence of a break-up event in Rodinia around 820–750 Ma (Li et al., 2003b). A comparison of paleolatitudes at reference points in Tarim, South China and Australia shows that the three blocks occupied the equatorial tropical latitudes during around 750 Ma to the early Paleozoic (Fig. 8), suggesting that the Tarim Block could have remained in close proximity to Western Australia during the late Neoproterozoic and the early Paleozoic. The close paleolatitudes of Tarim, South China and Western Australia between ca. 750 Ma and the early Paleozoic is coincident with their close Neoproterozoic tectonostratigraphic records (e.g. Wang, 1986; Li et al., 1996; Preiss, 2000; Wang and Li, 2003; Xiao et al., 2004; Xu et al., 2005) and close early Paleozoic trilobite affinities (e.g. Burrett and Richardson, 1980). Nevertheless, detailed relationships between Tarim, South China and Australia during the late Neoproterozoic and early Paleozoic can be further constructed until more precisely dated and high-quality paleomagnetic data are available; at present these are particularly sparse for the Neoproterozoic and early Paleozoic of Tarim.

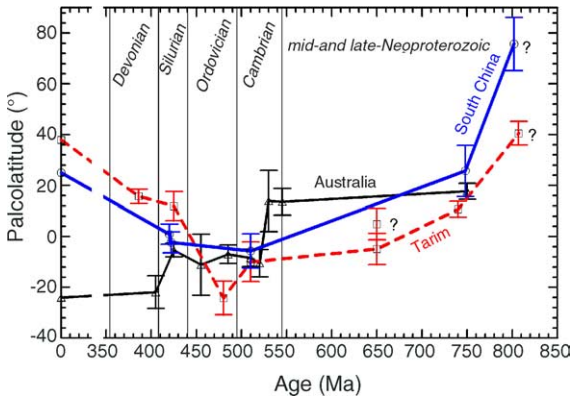


Fig. 8. Plot of paleolatitudes calculated from Neoproterozoic to early Paleozoic paleopoles of Tarim, South China and Australia at reference points of 38°N, 75°E (Tarim), 25°N, 100°E (South China) and of 24°S, 115°E (Australia). Vertical bars are latitudinal 95% confidence limits at the reference points. The Australian paleolatitudes are computed from a ca. 750 Ma pole (Wingate and Giddings, 2000) and an early Paleozoic APWP (van der Voo, 1993). Paleolatitudes for South China are computed from a ca. 800 Ma paleopole of Li et al. (2004) and from a ca. 750 Ma to early Paleozoic APWP after Yang et al. (2004), whereas paleolatitudes for Tarim are computed from a ca. 800 Ma pole of Chen et al. (2004), ca. 740 Ma pole of this study, a preliminary mean pole for mid- to late-Neoproterozoic rocks in Wushi (Li et al., 1991), and from an early Paleozoic APWP compiled by Fang et al. (1998).

Geological observations provide additional information supporting the view that the Tarim Block was most likely situated against Western Australia during most of the Neoproterozoic. There is a consensus that the 807–785 Ma mafic dyke swarm in the northwestern Tarim Block (Chen et al., 2004; Zhang L.F., personal communication) and the 783 ± 10 Ma bimodal intrusive complex in the southwestern Tarim Block (Zhang et al., in revision) are significant evidence that the Tarim Block was involved in mantle plume activity beneath west Rodinia possibly resulting in the break-up of the supercontinent (Li et al., 2003b). Meanwhile SHRIMP U-Pb zircon age and geochemical features of the Baiyisi volcanic rocks suggest that the Baiyisi glaciation could correlate with the Chang'an glaciation in the SCB and the Sturtain-Rapitan glaciation in other Rodinia blocks, and also suggesting that the Baiyisi bimodal volcanism was part of second global magmatism during the break-up of Rodinia (Xu et al., 2005). These observations indicate that the Tarim Block may be a candidate for a continental fragment that rifted from Western Australia during plate reorganization associated with the break-up of Rodinia at ca. 820–750 Ma (Li et al., 2003b). A preliminary mean paleopole obtained from the mid- and late-Neoproterozoic rocks in Wushi at the northwestern margin of the Tarim Block that unconformably overlies the early Neoproterozoic Aksu Group (Gao et al., 1985;

Li et al., 1991) indicates a possible southward drift of the Tarim Block relative to Western Australia (Fig. 8), which is agreement with sinistral shearing along the Darling mobile belt of the Southwestern Australia (Powell and Pisarevsky, 2002). Moreover, rift volcanics at several horizons throughout the mid- and late-Neoproterozoic succession in Quruktagh (e.g. Gao et al., 1985; Brookfield, 1994; Xiao et al., 2004) confirm that successive rifting occurred at the now northeastern margin of the Tarim Block (Brookfield, 1994).

5. Conclusions

A new paleomagnetic pole has been obtained from the SHRIMP U-Pb zircon dated Baiyisi volcanic rocks in the Xishankou area of the northeastern edge of the Tarim Block. Rock magnetic and microscopic observations indicate that the ChRM preserved in SD or PSD low-titanium titanomagnetite grains could be primary remanence. The new ca. 740 Ma paleomagnetic pole indicates that the Tarim Block was located in equatorial paleolatitudes supportive of low latitudes for the Sturtian glaciations and for the hypothesis of a snowball earth at ca. 750–720 Ma. Geological information and a comparison of ca. 750 Ma to early Paleozoic paleolatitudes in Tarim, South China and Australia suggest that the Tarim Block could be a continental fragment rifted from Western Australia during continental break-up. We argue that the Tarim Block was most likely situated close to South China and Western Australia in the early Neoproterozoic supercontinent and then drifted to equatorial paleolatitudes after the putative break-up event at ca. 820–750 Ma. However, further high-quality paleomagnetic data and geochronologic constraints are essential to perfect and refine the Neoproterozoic geodynamics of the Tarim Block.

Acknowledgements

We thank Drs. Zhong Zheng, Jinghui Guo for sparkling discussions and acknowledge Drs Zhengxiang Li, Lifei Zhang and Chuanlin Zhang for providing new geochronologic data from the Tarim Block. We also appreciate the assistance of Yongcheng Wang in the laboratory. Careful reviews by Dr. J.D.A. Piper and two anonymous reviewers improved the manuscript. This work is supported by the NSFC (Nos. 40032010 and 40221402). B. Huang acknowledges the Royal Society of the UK for supporting his visiting to the University of Liverpool.

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